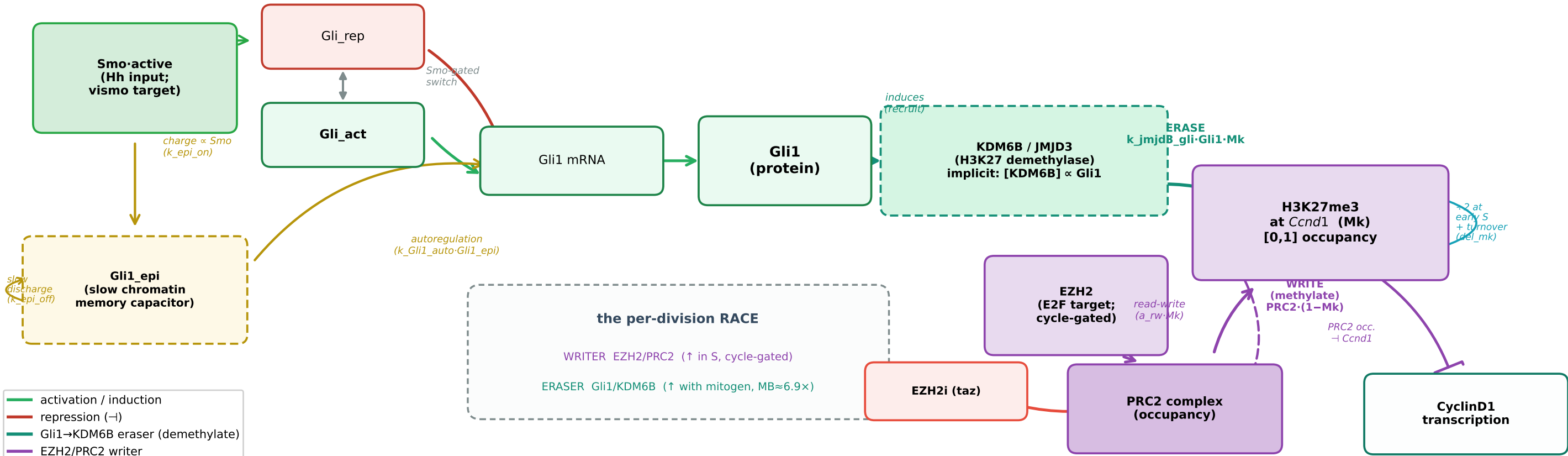


Gli1 → Kdm6b/Jmjd3 eraser of H3K27me3 at *Ccnd1* — detailed wiring + equations (v44.1)

mitogen-gated ERASER (Gli1 → KDM6B/JMJD3 demethylase) vs cycle-gated WRITER (EZH2/PRC2) — the per-division race that sets the mark



① Gli1 axis (mitogen input, with slow autoregulatory memory)

$$\begin{aligned} \text{Gli_rep} &\rightleftharpoons \text{Gli_act} && \text{(Smo-gated activator/repressor switch)} \\ d[\text{Gli1_mRNA}]/dt &= V_{\text{max}} \cdot \text{Gli_act}^2 / (K_{\text{GA}}^2 + \text{Gli_act}^2) \cdot (1 - \text{Gli_rep}^2 / (K_{\text{GR}}^2 + \text{Gli_rep}^2)) \cdot \gamma_{\text{Ptch}} \\ &\quad + k_{\text{Gli1_auto}} \cdot \text{Gli1_epi} - k_{\text{mdeg}} \cdot \text{Gli1_mRNA} \\ d[\text{Gli1_epi}]/dt &= k_{\text{epi_on}} \cdot (1 - \text{Ptch}) \cdot \text{Smo}^4 / (K_{\text{auto}}^4 + \text{Smo}^4) \cdot (1 - \text{Gli1_epi}) - k_{\text{epi_off}} \cdot \text{Gli1_epi} \\ d[\text{Gli1}]/dt &= k_{\text{Gli1_tl}} \cdot \text{Gli1_mRNA} - k_{\text{Gli1_deg}} \cdot \text{Gli1} \end{aligned}$$

② Eraser (KDM6B/JMJD3, implicit)

KDM6B is NOT a state variable.
Quasi-static: [KDM6B] ∝ Gli1,
folded into *k_jmjd3_gli*.

$$\begin{aligned} \text{erase flux} &= \\ & k_{\text{jmjd3_gli}} \cdot \text{Gli1} \cdot \text{Mk} \end{aligned}$$

③ H3K27me3 (Mk) balance at the *Ccnd1* domain

$$\begin{aligned} d[\text{Mk}]/dt &= \underbrace{\text{PRC2} \cdot (1 - \text{Mk})}_{\text{(EZH2/PRC2, cycle)}} - \underbrace{\text{del_mk} \cdot \text{Mk}}_{\text{(Gli1/KDM6B, mitogen)}} - \underbrace{k_{\text{jmjd3_gli}} \cdot \text{Gli1} \cdot \text{Mk}}_{\text{ERASE}} \\ \text{discrete: at Dna} &> 0.05 \rightarrow \text{Mk} \leftarrow \frac{1}{2} \cdot \text{Mk} && \text{(replicative dilution, once per S)} \\ \text{PRC2} &= \text{EZH2} \cdot (1 - \text{EZH2i}) \cdot (a_0 + a_{\text{rw}} \cdot \text{Mk}) \cdot (1 - g \cdot \text{Cd_mRNA}^p / (K_{\text{tx}}^p + \text{Cd_mRNA}^p)) \end{aligned}$$

④ Baked parameters (v44.1) + net effect

k_jmjd3_gli 0.0478 *del_mk* 0.0015
a0 3.78e-4 *a_rw* 4.30e-3 (*a_rw/a0* 11.4)
g_prc2 0.0424 *K_prc2* 3.5e-3 *n* 3.39
f0_prc2 0.0505 *K_tx* 5.12 *p_tx* 3.27
Gli: *Vmax* 1.27 *auto* 0.012 *on/off* 0.02/8e-4

MB *Gli1* ≈ 6.9× GNP → erase ≈ 6.9× stronger → ChIP *Ccnd1* mark MB/GNP ≈ 0.55 (MB < GNP)

Net over divisions: *Mk* accumulates when *WRITE* > (*turnover* + *ERASE* + ½-dilution) each cycle; MB's high *Gli1* tilts the race toward erasure despite higher *EZH2*.